

Response to Reviewer 2 - IoT-4522779

Manuscript IoT-4522779, MDPI *IoT*

Title: Modeling Normal Is All You Need: Joint Latent Clustering for Anomaly Detection in Multimodal Industrial-IoT Cyber-Physical Systems

Authors: Alexander Apartsin and Yehudit Aperstein

Dear Reviewer,

Thank you for reviewing the manuscript. Your suggestions on the abstract, introduction, related work, figures, notation, and typesetting are adopted, and the revised paper follows them closely. We address each point below, quoting your comment and then giving the change and its location. New and changed text is highlighted in green in the revised manuscript; deleted passages are removed.

Comment 1. Expand each abbreviation at first use (LatAD, IIoT, CPS, SCADA, and so on).

When an abbreviation first appears in the abstract or main text, its full form should be provided, e.g., LatAD, IIoT, CPS, SCADA, etc.

Response: change made. LatAD is given its full form at first use in the abstract ("Our detector, LatAD (Latent-density Anomaly Detector)"), and CPS, IIoT, and SCADA are expanded in Section 1 ("A cyber-physical system (CPS)"; "Industrial Internet of Things (IIoT)"; "Supervisory Control and Data Acquisition (SCADA) historians"). The remaining abbreviations (VaDE, GMM, NLL, HC, BIC, AUROC, F1, ICS, SOTA) are expanded at first use in the main text (for example Gaussian mixture model (GMM), Kullback–Leibler (KL), and multilayer perceptron (MLP)); the abstract keeps the compact forms and the main text re-expands them.

Comment 2. The abstract spends most of its length on the problems of existing methods; highlight LatAD's technical contribution.

The abstract spends most of its length introducing the problems of existing anomaly detection methods, while providing very little introduction to the principle of the paper's core contribution, LatAD. Please highlight the technical contributions of this paper in the abstract.

Response: change made. We agree. The abstract states LatAD's principle directly: "Our detector, LatAD (Latent-density Anomaly Detector), learns a latent Gaussian-mixture clustering of regimes (VaDE, Variational Deep Embedding), scores anomalies by density, and factorizes that density over correlation-community subsystems combined by a cohesion-weighted Higher Criticism statistic." It then reports the headline result: LatAD attains the best All-subset AUROC of any learned detector on HAI (0.948) and a significant lead on the difficult subset (HAI 0.845; the deep detectors USAD and TranAD fall to 0.44 to 0.50 on these reconstructable-but-improbable faults).

Comment 3. The introduction logic is unclear; delete the first-paragraph method sentence, keep contributions out of paragraph 1, and add CPS background.

The logic of the Introduction is not clear enough and should be carefully revised. In the first paragraph, the last sentence, "This paper detects them by XXX," should be deleted. The paper's contributions should not be introduced in the first paragraph of the Introduction. It is suggested to add background on cyber-physical systems in the first paragraph.

Response: change made. The "This paper detects them by ..." sentence is removed, no contributions appear in the first paragraph, and Section 1 now opens with CPS and IIoT background: "A cyber-physical system (CPS) couples networked sensors, actuators, and controllers with a physical process, forming the sensing-and-actuation layer of the Industrial Internet of Things (IIoT); monitoring its telemetry for faults and attacks is central to reliability and safety." The contributions are stated later in Section 1, after the problem and the two obstacles are set out.

Comment 4. Page 2, line 71: "the task is often framed as ..." - state which task.

On Page 2, line 71, "the task is often framed as XXX"-what task does "the task" specifically refer to? Please state it clearly.

Response: change made. The sentence now names the task explicitly (Section 1): "Unsupervised anomaly detection for CPS telemetry is often framed as characterizing the anomalies, but that framing is misleading."

Comment 5. The "two obstacles": does LatAD contribute to both, and what problem does it address or improve?

The paper states that modeling and evaluation are "two obstacles that have obscured this failure mode." Has LatAD made contributions to both aspects? What specific problem is LatAD proposed to address, or what existing method does it improve? Please explain clearly.

Response: change made. LatAD contributes to both obstacles, and Section 1 now states this through its three contributions. On the modeling obstacle, contribution C1 replaces the reconstruction residual of USAD- and TranAD-style deep detectors with a probability score in a jointly learned latent regime space, factorized over the plant's physical subsystems. On the evaluation obstacle, contribution C2 is a detector-agnostic difficulty stratification, the channel-based rule and the linear principal-component subspace filter that isolate the difficult and double-hard subsets; the underlying raw-metric protocol it builds on, point-wise scoring with no point adjustment, is adopted from the critique rather than claimed as novel. Contribution C3 adds a drift-aware anomaly typing, so LatAD contributes on both the modeling and the evaluation axes.

Comment 6. Restructure Related Work: move the anomaly-detection definition to the introduction, delete the three LatAD self-descriptions, rewrite Section 2.3, and delete Section 2.5.

The structure of the Related Work section also has major problems and should be revised. First, the definition of anomaly detection in the first sentence should be moved to the Introduction. Second, since LatAD is the authors' novel work, it should not be described in detail in Related Work. It is suggested to delete: the last sentence of the first paragraph, "LatAD sits squarely in this line XXX"; the last sentence of Section 2.1, "LatAD addresses a complementary IIoT problem XXX"; and the last paragraph of Section 2.2, "LatAD adopts this critique as XXX." Section 2.3 is entirely an introduction to LatAD and needs to be rewritten. Section 2.5 is suggested to be deleted.

Response: change made. The anomaly-detection definition is relocated to Section 1. The three LatAD self-descriptions (the last sentence of the Related Work preamble, the last sentence of Section 2.1, and the last paragraph of Section 2.2) are removed. Section 2.3 is rewritten as a neutral survey of latent and clustering models and now cites Deep SVDD, OmniAnomaly, and THOC. The former Section 2.5 "Benchmark datasets" is removed; in its place Section 2.5 is now "Benchmark versus operational scale", a short scholarly paragraph on the size of public CPS benchmarks versus operational scale, which supports the scoping of assumptions A8 to A10 (a pointer to Section 5.1 covers the dataset details).

Comment 7. Add a system-model or application-scenario diagram showing where in the IIoT LatAD collects data and detects.

Add a system model diagram or an application scenario diagram that shows at which stage of the Industrial IoT the proposed LatAD method collects data and performs anomaly detection.

Response: change made. We agree. Figure 1 (Section 1) shows the IIoT monitoring context: field sensors and PLCs feed an edge gateway or SCADA historian that aggregates and windows the multivariate telemetry, LatAD scores each completed window online, and an anomaly alarm together with the most-surprised subsystem is forwarded to the operator, maintenance, or security stack. It is referenced from Section 1, and the caption notes that model fitting and calibration use train-normal telemetry only, so a site is commissioned from its own normal data without labeled attacks. The most-surprised community Figure 1 forwards is now an evaluated triage output: Section 8 (new Table 5) reports that a three-community shortlist ranked by train-normal p-value contains the attacked subsystem in 68 percent of HAI and 56 percent of WADI attacks, against 56 and 31 percent for a random shortlist of the same size.

Comment 8. Explain the relationship between MIIM (Section 3) and the LatAD detector (Section 4).

What is the relationship between the MIIM presented in Section 3 and the LatAD detector presented in Section 4? Are they studied independently or closely related? Please add an explanation of their relationship.

Response: change made. Section 3 now adds a checklist sentence stating that MIIM is a specification a dataset realizes to varying degrees, and Section 4 states the relationship explicitly, immediately after the workflow figure: "A1 to A10 specify what a model of CPS normal must capture, not how to implement it; the detector below is one concrete realization of A1 to A7." Section 4 then maps every realized assumption (A1 to A7) to the LatAD design section and mechanism that implements it, and Table 1 lists the ten assumptions with their physical origin. The caption states which assumptions are realized (A1 to A7) and which are specified for future work (A8 to A10), so the mapping is stated explicitly.

Comment 9. Section 4 readability: map Sections 4.1 to 4.4 to pipeline stages and add a LatAD workflow figure.

The readability of Section 4 needs to be improved. Sections 4.1 to 4.4 barely mention LatAD in the main text, making it unclear which part of the LatAD detector pipeline each corresponds to. Which parts of the LatAD detector pipeline do the representation stage and scoring stage correspond to? Please add a figure to introduce the LatAD workflow.

Response: change made. Each Section 4 heading now names its pipeline stage: Section 4.1 "Representation: Variational Deep Embedding (VaDE) (the representation stage; Figure 2, top)"; Section 4.3 "The scoring stack (the scoring stage; Figure 2, per-community heads)"; and Section 4.4 "Regime-community density (the headline realization and subsystem-factorization stage; Figure 2, community fan-out and HC combiner)". Figure 2, the LatAD workflow with each block tagged by the assumption it serves, is added and referenced from the Section 4 preamble.

Comment 10. Is VaDE plus density plus residual fitted separately per community, or is the encoder shared? Report training time, GPU memory, and inference latency.

Is VaDE + density head + residual head fitted separately for each community? Or is the encoder shared? The paper does not report training time, GPU memory, or inference latency, which is critical for IIoT edge deployment.

Response: change made. Both are now answered directly. Section 4.4 states the architecture: for each community we fit a separate small VaDE, each with its own encoder, Gaussian mixture, and scoring heads, its regime count and latent dimension scaled to the community size; nothing is shared across communities except the input window features, so the model is a stack of independent per-subsystem experts plus one global expert. The encoder is therefore not shared. Cost is reported in Appendix B and Table B1: parameter

count, training wall-clock, per-window inference latency, peak GPU memory, and host memory for LatAD (global and regime-community) and the deep baselines, on one shared GPU machine plus a CPU edge proxy. LatAD-global trains in 3.2 / 16 / 3.5 s on WADI / HAI / SWaT (roughly 47 to 72 times faster than TranAD), per-window latency is 0.7 to 13 ms global and 17 to 32 ms for the full community stack, and both configurations run within about 3.5 to 3.9 GB of host memory.

Comment 11. The symbol z denotes both a latent variable (Section 4.3) and a standardized per-channel value (Section 5.3); replace one.

The symbol z denotes a latent variable in §4.3 but a standardized per-channel value in §5.3. Although the paper includes the sentence "here z is a standardized per-channel value, not the latent z of §4.3," the symbol conflict is still serious. It is suggested to replace z with another symbol in §4.2 or §5.3.

Response: change made. The conflict is resolved: the latent variable keeps the symbol z , the standardization operator is renamed $\text{std}(\cdot)$, and the trivial-rule per-channel value is renamed u throughout Table 2 (and Table A3) and the prose (Sections 5.3 and 5.5, Section 6). The old disambiguation clause is replaced by a short parenthetical at first use in Section 5.3, and the remaining "z-score units" phrasing in Section 4.4 and the Table A1 caption is changed to "standardized units."

Comment 12. Typesetting: unify paragraph indentation and number all equations.

Typesetting issues: Some paragraphs begin flush left, while others are indented. Please unify the format. None of the equations in the paper are numbered.

Response: change made. All nine display equations are now numbered (1) to (9), right-aligned (Sections 3, 4.1, 4.3, and 4.4). Paragraph-indentation is unified under one body style, applied at the DOCX build stage so that every paragraph follows the same first-line-indent convention.

On results clarity, English, and figures.

The results should be clearer, the English could be improved, and the figures and tables should be well-presented.

Response: change made. The results section was rewritten around the corrected, construct-matched numbers, with the significance test and window and episode counts stated for each dataset, and a new source-of-gain table (Table A4) added. A construct-matched graph baseline (GDN) was added to Table 2 and to Figure 3. Figures 1 and 2 were added and the results chart renumbered Figure 3 with a corrected caption. We revised the front matter and the longer sentences, and a language pass over the whole manuscript is applied to the submitted version.

Thank you again for reviewing the manuscript; the revised paper follows the suggestions throughout.

Sincerely,
Alexander Apartsin and Yehudit Aperstein